

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES									
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.								
20										817	USE ONLY FOR TIE IN								
20	U04		MODULAR DBB	K03AC02	0.75	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.								
30	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B1	ALL BOLTS ARE METRIC DIA.								
40	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10								
50	U04	VAW	MODULAR DBB	M03EC02	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22								
60							***** BALL VALVES *****			FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES								
60		VB2								FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.								
70	U04		BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0										
80	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0										
90	U04		BALL	A03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0										
100	U04		BALL	A03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	NT5	TO BE USED IN NON-PIGGED LINES ONLY								
110	U04	VA5	BALL	B03AC01	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	SB6	USE JACKSCREWS FOR PADDLE								
120	U04	VA5	BALL	B03AC02	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.								
130	U04	VA5	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0										
140	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS								
150	U04	VA5	BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0										
160	U04	VA5	BALL	B03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.								
170	U04	VA5	BALL	B03JC07	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0										
180	U04	VA5	BALL	B03JC08	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.								
190							***** RISING STEM BALL VALVES *****			VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.								
200	U04	VA5	BALL	R03AC06	2	8	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0										
210		VA5	BALL	R03AC08	6	10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0										
220							***** BUTTERFLY VALVES *****												
220																			
230			BUTTERFLY	T03AC61	10	24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0										
240			BUTTERFLY	T03JC61	28	28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0										
250							***** GATE VALVES *****												
250																			
260																			
260	U04		GATE	W03AC70	3	6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0										
270	U04		GATE	W03AC71	8	24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0										
280			GATE	W03JC71	28	28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A	0	0										
290							***** CHECK VALVES *****												
290																			
300																			
300	U04	VC1	CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0										
310	U04		CHECK	C03AC45	2	10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0										
320	U04		CHECK	C03AC52	12	24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16	0	0										
330	U04		CHECK	C03JC52	28	28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0										
340							***** GLOBE VALVES *****												
340																			
350																			
350	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0										
360			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0										
370			GLOBE	L03AC42	8	10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0										
380							***** PIPE ABOVE GROUND *****												
380																			
390																			
390			PIPE	5358512	0.75	0.75	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0										
400			PIPE	5358508	1	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0										
410			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0										
420			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0										
430																			
430			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0										
440			PIPE	9001254	6	10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0										
450			PIPE	9001256	12	14	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE	0	0										
460			PIPE	9001311	16	18	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0										
470			PIPE	9001485	20	20	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0										
480			PIPE	9001493	24	24	LINEPIPE SMLS WT LS WT = 9.53 MM API-5L X60 BE	0	0										
490			PIPE	9001494	28	28	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE	0	0										
500																			
500		812	PIPE	5358513	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0										
510																			
			TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS		
					ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO		MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C								

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES																				
510	U04	821	PIPE	9005244	16	16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0																					
520	U04	821	PIPE	9005242	18	18	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE	0	0																					
530	U04	821	PIPE	9001356	16	16	LINEPIPE SMLS WT = 30.96 MM API-5L X60 BE	0	0																					
540	U04	821	PIPE	9001357	18	18	LINEPIPE SMLS WT = 34.93 MM API-5L X60 BE	0	0																					
550	U04	821	PIPE	9001495	8	16	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE	0	0																					
560	U04	821	PIPE	9005243	20	20	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE	0	0																					
570	U04	821	PIPE	9005254	24	24	LINEPIPE SMLS WT= 20.62 MM API-5L X60 BE	0	0																					
580																														
580							***** PIPE BELOW GROUND *****																							
590																														
590			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0																					
600			PIPE	9001254CTD	6	10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0																					
610			PIPE	9001256CTD	12	14	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED	0	0																					
620			PIPE	9001311CTD	16	18	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0																					
630			PIPE	9001485CTD	20	20	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0																					
640			PIPE	9001493CTD	24	24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0																					
650			PIPE	9001494CTD	28	28	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED	0	0																					
660							***** FITTINGS ABOVE & BELOW GROUND*****																							
660																														
670																														
670			ELBOW	5392816	0.75	0.75	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0																					
680			ELBOW	5392600	1	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0																					
690			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0																					
700			ELBOW	9006566	4	4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60	0	0																					
710			ELBOW	9002190	6	10	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0																					
720			ELBOW	9006477	12	14	ELL BW 90 DEG LR WT 7.14 MM A860 WPHY 60	0	0																					
730			ELBOW	9006478	16	18	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60	0	0																					
740			ELBOW	9006481	20	20	ELL BW 90 DEG LR WT 8.74 MM A860 WPHY 60-W	0	0																					
750			ELBOW	9006472	24	24	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60-W	0	0																					
760			ELBOW	9006473	28	28	ELL BW 90 DEG LR WT 10.31 MM A860 WPHY 60-W	0	0																					
770	U04	821	ELBOW	9006479	16	16	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60	0	0																					
780	U04	821		9006469	16	16	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60	0	0																					
790			ELBOW	5392822	0.75	0.75	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0																					
800			ELBOW	5392606	1	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0																					
810			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0																					
820			ELBOW	9006571	6	10	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0																					
830			ELBOW	9006589	12	14	ELL BW 45 DEG LR WT 7.14MM A860 WPHY 60	0	0																					
840			ELBOW	9006588	16	18	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60	0	0																					
850			ELBOW	9006570	24	24	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60-W	0	0																					
860	U04		ELBOW	9006573	4	4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60	0	0																					
870	U04		ELBOW	9006593	20	20	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60	0	0																					
880																														
880			TEE	5392825	0.75	0.75	TEE BW SCH 160 LTCS A420 WPL6	0	0																					
890			TEE	5392609	1	2	TEE BW SCH XS LTCS A420 WPL6	0	0																					
900			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0																					
910		NT5	TEE	9301610	4	4	TEE BW WT 6.02 MM LTCS A860 WPHY-60	0	0																					
920		NT5	TEE	9301609	6	10	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0																					
930		NT5	TEE	9301605	12	14	TEE BW WT 7.14 MM LTCS A860 WPHY-60	0	0																					
940		NT5	TEE	9301606	16	18	TEE BW WT 7.92 MM LTCS A860 WPHY-60	0	0																					
950		NT5	TEE	9006487	20	20	TEE BW WT 8.74MM A860 WPHY 60-W	0	0																					
960		NT5	TEE	9004210	24	24	TEE BW WT 9.53MM A860 WPHY 60-W	0	0																					
970	U04	821	TEE	9006688	24	24	TEE BW HEADER WT 20.62MM/BRANCH 20.62MM Tapered to 9.53MM A860 WPHY 60-W	0	0																					
980	U04	821	TEE	9006633	16	16	TEE BW WT 12.7 MM A860 WPHY 60	0	0																					
990	U04	821	TEE	9006683	20	20	TEE BW WT 12.7MM Taper Bored to 8.74MM /12.7 MM Taper bored to 8.74MM LTCS A860 WPHY 60	0	0																					
1000	U04	821	TEE	9004200	16	16	TEE BW WT 15.88MM A860 WPHY 60	0	0																					
1010	U04	821	TEE	9006682	16	16	TEE BW WT 15.88MM Taper Bored to 7.92MM /15.88 MM Taper bored to 7.92MM LTCS A860 WPHY 60	0	0																					
1020	U04	821	TEE	9004201	18	18	TEE BW WT 17.48MM A860 WPHY 60	0	0																					
1030	U04	821		9004212	20	20	TEE BW WT 19.05MM A860 WPHY 60-W	0	0																					
1040	U04	821	TEE	9004213	20	24	TEE BW WT 20.62MM A860 WPHY 60-W	0	0																					
1050	U04	NT5	TEE	9301612	24	24	TEE BW WT 9.53 MM LTCS A860 WPHY-60	0	0																					
1060		NT5	TEE RED	5392469	0.75	0.75	TEE RED BW SCH XS/160 LTCS A420 WPL6	1	1.5																					
1070		NT5	TEE RED	5392661	1	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2																					
1080		NT5	TEE RED	5392454	1	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3																					
1090		NT5	TEE RED	9006664	2	2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4	4																					
1100		NT5	TEE RED	9006659	3	3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4	4																					
1110		NT5	TEE RED	9006658	3	3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6	6																					
1120		NT5	TEE RED	9006605	4	4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6	10																					
1130		NT5	TEE RED	9006014	6	8	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60	8	10																					
1140		NT5	TEE RED	9006015	6	10	TEE RED BW WT 7.14 MM/6.35 MM LTCS A860 WPHY 60	12	14																					
1150		NT5	TEE RED	9006637	12	12	TEE RED BW WT 7.14 MM LTCS A860 WPHY 60	14	14																					
1160			TEE RED	9006017	6	6	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16	16																					
1170		NT5	TEE RED	9006017	8	10	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16	18																					
1180		NT5	TEE RED	9006016	12	14	TEE RED BW WT 7.92 MM/7.14 MM LTCS A860 WPHY 60	16	18																					
<table><tr><td rowspan="2"></td><td rowspan="2">TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN</td><td>CONSTRUCTION</td><td>CORR.ALLOW.</td><td>SERVICE</td><td>MATERIAL</td><td>DATE</td><td>BY</td><td>REV</td><td>SHEET</td><td>PIPING LINE CLASS</td></tr><tr><td>ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE</td><td>CORR. ALLOW. 3 MM</td><td>SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO</td><td>MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C</td><td>12/16/2024</td><td>BY OZAC</td><td>U04</td><td>Page 2 of 6</td><td>300K5A</td></tr></table>												TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS	ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 2 of 6	300K5A
	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS																				
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1190	U04	NT5	TEE RED	9006018	16	16	TEE RED BW WT 7.92 MM LTCS A860 WPHY 60	18	18						
1200			TEE RED	9004143	8	8	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W	20	20						
1210		NT5	TEE RED	9004143	10	10	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W	20	20						
1220	U04	NT5	TEE RED	9006020	12	14	TEE RED BW WT 8.74 MM/7.14 MM LTCS A860 WPHY 60	20	20						
1230	U04	NT5	TEE RED	9006019	16	18	TEE RED BW WT 8.74 MM/7.92 MM LTCS A860 WPHY 60	20	20						
1240		NT5	TEE RED	9004149	10	10	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	24	24						
1250	U04	NT5	TEE RED	9006606	12	14	TEE RED BW WT 9.53 MM/7.14 MM LTCS A860 WPHY 60	24	24						
1260		NT5	TEE RED	9006483	16	18	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60-W	24	24						
1270	U04	NT5	TEE RED	9006484	20	20	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60	24	24						
1280		NT5	TEE RED	9006635	20	20	TEE RED BW WT 10.31 MM/8.74 MM LTCS A860 WPHY 60-W	28	28						
1290		NT5	TEE RED	9006636	24	24	TEE RED BW WT 10.31 MM/9.53 MM LTCS A860 WPHY 60-W	28	28						
1300	U04	821		9006692	12	12	TEE RED BW 12.7 MM Taper Bored to 8.74 /12.7 MM LTCS A860 WPHY 60-W	20	20						
1310	U04	821	TEE RED	9002328	10	10	TEE RED BW SCH 120 Taper bored to 7.92 /WT 21.44MM LTCS A860 WPHY 60	16	16						
1320	U04	821	TEE RED	9006681	6	6	TEE RED BW WT 12.7 MM/6.35 MM LTCS A860 WPHY 60	16	16						
1330	U04	821	TEE RED	9008862	12	12	TEE RED BW WT 12.7 MM/7.11MM LTCS A860 WPHY 60	16	16						
1340	U04	821	TEE RED	9006689	18	18	TEE RED BW WT 12.7 Tapered to 9.53MM MM/12.70 MM LTCS A860 WPHY 60	24	24						
1350	U04	821		9006693	12	12	TEE RED BW WT 12.70 MM/ WT, TAPERED TO 7.9MM/ 12.7 MM LTCS A860 WPHY 60	16	16						
1360	U04	821		9006695	6	6	TEE RED BW WT 15.88 MM/6.35 MM LTCS A860 WPHY 60	16	16						
1370	U04	821		9006694	12	12	TEE RED BW WT 15.88MM /15.88 MM LTCS A860 WPHY 60	16	16						
1380	U04	821	TEE RED	9006656	16	16	TEE RED BW WT 17.48 MM/15.88 MM LTCS A860 WPHY 60	18	18						
1390	U04	821	TEE RED	9002332	12	12	TEE RED BW WT 17.48MM Taper bored to 7.92 /12.7 MM LTCS A860 WPHY 60	18	18						
1400	U04	821	TEE RED	9002330	16	16	TEE RED BW WT 17.48MM Taper bored to 7.92 /15.88 MM LTCS A860 WPHY 60	18	18						
1410	U04	821	TEE RED	9002331	16	16	TEE RED BW WT 17.48MM Taper bored to 7.92 MM /15.88 MM Taper bored to 7.92 MM LTCS A860 WPHY 60	18	18						
1420	U04	821	TEE RED	9006684	12	12	TEE RED BW WT 19.05MM Taper Bored to 8.74MM /18.26MM LTCS A860 WPHY 60	20	20						
1430	U04	821		9006690	20	20	TEE RED BW WT 20.62 MM /19.05 Taper Bored to 8.74MM LTCS A860 WPHY 60-W	24	24						
1440	U04	821	TEE RED	9006686	18	18	TEE RED BW WT 20.62 MM Taper Bored to 9.53 /17.48 MM LTCS A860 WPHY 60-W	24	24						
1450	U04	821	TEE RED	9006687	20	20	TEE RED BW WT 20.62 MM Taper Bored to 9.53 /19.05 tapered to 8.74 MM LTCS A860 WPHY 60-W	24	24						
1460	U04	821		9006691	18	18	TEE RED BW WT 20.62 MM Taper Bored to 9.53MM /17.48 Tapered to 7.92 MM LTCS A860 WPHY 60-W	24	24						
1470	U04	821	TEE RED	9004204	18	18	TEE RED BW WT 20.62 MM/17.48 MM LTCS A860 WPHY 60-W	24	24						
1480	U04	821		9004205	20	20	TEE RED BW WT 20.62 MM/19.05 MM LTCS A860 WPHY 60-W	24	24						
1490	U04	821	TEE RED	9002329	12	12	TEE RED BW WT 34.93MM Taper bored to 7.92 MM/25.4 MM LTCS A860 WPHY 60	18	18						
1500			CAP	5392826	0.75	0.75	CAP BW SCH 160 LTCS A420 WPL6	0	0						
1510			CAP	5392610	1	2	CAP BW SCH XS LTCS A420 WPL6	0	0						
1520			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0						
1530			CAP	9005055	6	10	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0						
1540			CAP	9005057	12	14	CAP BW WT 7.14 MM LTCS A860 WPHY-60	0	0						
1550			CAP	9005058	16	18	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0						
1560			CAP	9005059	20	22	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0						
1570			CAP	9005060	24	24	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0						
1580			CAP	9005061	28	28	CAP BW WT 10.31 MM LTCS A860 WPHY-60	0	0						
1590															
1590			RED CONC	5392462	0.75	0.75	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1	2						
1600			RED CONC	5392458	1	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1.5	2						
1610			RED ECC	5392465	0.75	0.75	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1	2						
1620			RED ECC	5392461	1	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1.5	2						
1630			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3						
1640			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3						
1650			RED CONC	9006918	1.5	3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4						
1660			RED ECC	9001372	1.5	3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4						
1670			RED CONC	9006915	3	3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6	6						
1680	U04		RED ECC	9002510	3	3	REDUCER ECC WT 6.35 MM/STD LTCS A860 WPHY-60	6	6						
1690			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	10						
1700			RED ECC	9002266	6	8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	10						
1710			RED ECC	9002269	8	10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12	14						
1720			RED ECC	9006908	12	12	REDUCER ECC WT 7.14 MM LTCS A860 WPHY-60	14	14						
1730			RED ECC	9002272	10	10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16	18						
1740			RED ECC	9002270	12	14	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60	16	18						
1750			RED ECC	9002278	16	16	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	18	18						
1760	U04	821	RED ECC	9002405	8	8	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60	12	12						
1770	U04	821		9006936	8	8	REDUCER ECC WT 12.70MM / 12.70 MM LTCS A860 WPHY-60	12	12						
1780	U04	821	RED ECC	9006936	10	10	REDUCER ECC WT 12.70MM / 12.70 MM LTCS A860 WPHY-60	18	18						
1790	U04	821	RED ECC	9002274	8	10	REDUCER ECC WT 15.88 MM / 11.13 MM LTCS A860 WPHY-60	12	16						
1800	U04	821	RED ECC	9004282	10	10	REDUCER ECC WT 17.48 MM / 11.13 MM LTCS A860 WPHY-60	18	18						
1810	U04	821	RED ECC	9004283	12	12	REDUCER ECC WT 17.48 MM / 12.70 MM LTCS A860 WPHY-60	18	18						
1820	U04	821	RED ECC	9004312	8	8	REDUCER ECC WT 18.26MM LTCS A860 WPHY-60	12	12						
1830	U04		RED ECC	9002294	12	14	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60	20	20						
1840	U04		RED ECC	9002293	16	18	REDUCER ECC WT 8.74 MM / 7.92 LTCS A860 WPHY-60	20	20						
1850	U04		RED ECC	9002251	16	18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	24	24						
1860			RED ECC	9004132	12	14	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60-W	20	20						
1870			RED ECC	9004133	16	18	REDUCER ECC WT 8.74 MM / 7.92MM LTCS A860 WPHY-60-W	20	20						
1880			RED ECC	9004134	16	16	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	24	24						
	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS	
			ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE:- 40 / 88 °C		12/16/2024	BY OZAC	U04	Page 3 of 6	300K5A	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES					
1890	U04		RED CONC	9006901	16	18	REDUCER CONC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	24	24						
1900			RED ECC	9004135	16	18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPH Y-60-W	24	24						
1910	U04		RED CONC	9006902	20	20	REDUCER CONC WT 9.53 MM / 8.74MM LTCS A860 WPHY-60	24	24						
1920	U04	821	RED CONC	9006911	18	18	REDUCER CONC WT 20.62 MM / 17.48 MM LTCS A860 WPHY-60	24	24						
1930	U04	821	RED ECC	9002273	6	10	REDUCER ECC WT SCH 120 / SCH120 LTCS A860 WPHY-60	10	16						
1940	U04		RED CONC	9006931	12	12	REDUCER CONC WT 7.92 MM / 7.14 WT LTCS A860 WPHY-60	16	18						
1950	U04		RED CONC	9006928	18	18	REDUCER CONC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	20	20						
1960			RED ECC	9004136	20	20	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	24	24						
1970	U04		RED CONC	9006903	24	24	REDUCER CONC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60	28	28						
1980			RED ECC	9004137	24	24	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60-W	28	28						
1990							*****INDUCTION BENDS ABOVE GROUND*****								
1990															
2000															
2000			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2010			INDUC BEND	9001813	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2020			INDUC BEND	9001814	12	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2030			INDUC BEND	9001815	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2040			INDUC BEND	9001816	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2050			INDUC BEND	9001817	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2060			INDUC BEND	9001818	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2070			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2080			INDUC BEND	9001823	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2090			INDUC BEND	9001824	12	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2100			INDUC BEND	9001825	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2110			INDUC BEND	9001826	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2120			INDUC BEND	9001827	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2130			INDUC BEND	9001828	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2140			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2150			INDUC BEND	9001833	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2160			INDUC BEND	9001834	12	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2170			INDUC BEND	9001835	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2180			INDUC BEND	9001836	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2190			INDUC BEND	9001837	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2200			INDUC BEND	9001838	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0						
2210							*****INDUCTION BENDS BELOW GROUND*****								
2210															
2220															
2220			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2230			INDUC BEND	9001813CTD	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2240			INDUC BEND	9001814CTD	12	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2250			INDUC BEND	9001815CTD	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2260			INDUC BEND	9001816CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2270			INDUC BEND	9001817CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2280			INDUC BEND	9001818CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2290	U04		INDUC BEND	9007034CTD	18	18	INDUCTION BEND 90 DEG X 10D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2300	U04		INDUC BEND	9007035CTD	20	20	INDUCTION BEND 90 DEG X 10D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2310			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2320			INDUC BEND	9001823CTD	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2330			INDUC BEND	9001824CTD	12	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2340			INDUC BEND	9001825CTD	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2350			INDUC BEND	9001826CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2360			INDUC BEND	9001827CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2370			INDUC BEND	9001828CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2380	U04		INDUC BEND	9007004CTD	18	18	INDUCTION BEND 45DEG X 10D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2390	U04		INDUC BEND	9007005CTD	20	20	INDUCTION BEND 45DEG X 10D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2400	U04		INDUC BEND	9007008CTD	24	24	INDUCTION BEND 45DEG X 10D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2410	U04		INDUC BEND	9007019CTD	18	18	INDUCTION BEND 45DEG X 15D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2420	U04		INDUC BEND	9007020CTD	20	20	INDUCTION BEND 45DEG X 15D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2430	U04		INDUC BEND	9007023CTD	24	24	INDUCTION BEND 45DEG X 15D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2440			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2450			INDUC BEND	9001833CTD	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2460			INDUC BEND	9001834CTD	12	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2470			INDUC BEND	9001835CTD	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2480			INDUC BEND	9001836CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
 TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS	
			ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C		12/16/2024	BY OZAC	U04	Page 4 of 6	300K5A	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES					
2490			INDUC BEND	9001837CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2500			INDUC BEND	9001838CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0						
2510							***** OLETS ABOVE & BELOW GROUND****								
2510															
2520															
2520			OLETS	9006400	0.75	0.75	WELDOLET WT 6.02 MM / SCH 160 LTCS A694 F60	4	4						
2530	U04		OLETS	9006219	1	1.5	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4						
2540			OLETS	9002450	0.75	0.75	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	6	10						
2550			OLETS	9006407	1	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	10						
2560			OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8	10						
2570			OLETS	9006817	0.75	0.75	WELDOLET WT 7.14 MM/ SCH 160 WT LTCS A694 F60	12	14						
2580			OLETS	9006413	1	2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12	14						
2590			OLETS	9006412	3	3	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12	14						
2600			OLETS	9006411	4	4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	12	14						
2610			OLETS	9006818	0.75	0.75	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	16	18						
2620			OLETS	9002552	1	2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	16	18						
2630			OLETS	9006415	3	3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	16	18						
2640			OLETS	9006414	4	4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	16	18						
2650			OLETS	9004124	6	6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	18	18						
2660			OLETS	9002455	0.75	0.75	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	20	20						
2670			OLETS	9006423	1	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	20	20						
2680			OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	20	20						
2690			OLETS	9006421	4	4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	20	20						
2700			OLETS	9002008	6	6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	20	20						
2710			OLETS	9002457	0.75	0.75	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	24	24						
2720			OLETS	9006429	1	2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	24	24						
2730			OLETS	9006428	3	3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	24	24						
2740			OLETS	9006427	4	4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	24	24						
2750			OLETS	9006424	6	8	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	24	24						
2760			OLETS	9006433	4	4	WELDOLET WT 10.31 MM/6.02 MM LTCS A694 F60	28	28						
2770			OLETS	9006430	6	10	WELDOLET WT 10.31 MM/6.35 MM LTCS A694 F60	28	28						
2780			OLETS	9006816	12	12	WELDOLET WT 10.31 MM/7.14 MM LTCS A694 F60	28	28						
2790	U04		OLETS	5504380	1	1	WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1	0	0						
2800	U04		OLETS	5505179	1	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	3						
2810	U04		OLETS	5505333	1	1	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	2						
2820	U04	821	OLETS	9002467	1	1	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	16	16						
2830	U04	821	OLETS	9005087	3	3	WELDOLET WT 12.70 MM/STD WT LTCS A694 F60	16	16						
2840	U04	821		9005083	3	3	WELDOLET WT 15.88 MM/ SCH STD LTCS A694 F60	16	16						
2850	U04	821		9005045	1	1	WELDOLET WT 15.88 MM/ SCH XS LTCS A694 F60	16	16						
2860	U04	821		9005126	2	2	WELDOLET WT 20.62 MM/ SCH XS LTCS A694 F60	24	24						
2870	U04		OLETS	9006408	4	4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	12	12						
2880															
2880	U04		OLETS	5504450	0.75	0.75	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	2	2						
2890	U04		OLETS	5504451	0.75	0.75	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2 CL.1	3	3						
2900			OLETS	9006807	0.75	0.75	NIPOFLANGE 300# RF 6.02 MM/SCH 160 WT LTCS A694 F60	4	4						
2910			OLETS	9006808	0.75	0.75	NIPOFLANGE 300# RF 6.35 MM/SCH 160 WT LTCS A694 F60	6	10						
2920			OLETS	9006809	0.75	0.75	NIPOFLANGE 300# RF 7.14 MM/SCH 160 WT LTCS A694 F60	12	14						
2930			OLETS	9006810	0.75	0.75	NIPOFLANGE 300# RF 7.92 MM/SCH 160 WT LTCS A694 F60	16	18						
2940			OLETS	9006811	0.75	0.75	NIPOFLANGE 300# RF 8.74 MM/SCH 160 WT LTCS A694 F60	20	20						
2950			OLETS	9006812	0.75	0.75	NIPOFLANGE 300# RF 9.53 MM/SCH 160 WT LTCS A694 F60	24	24						
2960			OLETS	9006813	0.75	0.75	NIPOFLANGE 300# RF 10.31 MM/SCH 160 WT LTCS A694 F60	28	28						
2970	U04		OLETS	5504020	1	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2						
2980	U04		OLETS	5504021	1	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3						
2990			OLETS	9002639	1	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4						
3000			OLETS	9002643	1	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	10						
3010			OLETS	9002644	1	2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12	14						
3020			OLETS	9002645	1	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	16	18						
3030			OLETS	9002646	1	2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	20	20						
3040			OLETS	9002647	1	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	24	24						
3050			OLETS	9002648	1	2	NIPOFLANGE 300# RF 10.31 MM/XS WT LTCS A694 F60	28	28						
3060	U04	821	OLETS	9002650	1	2	NIPOFLANGE 300# RF 12.7 MM/XS WT LTCS A694 F60	16	16						
3070	U04	821		9006828	1	2	NIPOFLANGE 300# RF 15.88 MM/XS WT LTCS A694 F60	16	16						
3080	U04	821	OLETS	9006827	1	1	NIPOFLANGE 300# RF 20.62 MM/ XS WT LTCS A694 F60	24	24						
3090	U04	821	OLETS	9006826	0.75	0.75	NIPOFLANGE 300# RF 20.62 MM/SCH 160 WT LTCS A694 F60	24	24						
3100	U04	FF5	OLETS	9504524	0.75	1	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A694 F60	0	0						
3110							***** FLANGES ABOVE & BELOW GROUND*****								
3110															
3120															
3120	U04		FLANGE	5556480	0.5	0.75	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 160	0	0						
3130	U04		FLANGE	5556476	1	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0						
3140	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0						
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS	
			ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C		12/16/2024	BY OZAC	U04	Page 5 of 6	300K5A	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3150			FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0	
3160			FLANGE	9001483	6	10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0	
3170			FLANGE	9001484	12	14	FLG WN 300# RF LTCS A694 F60 WT 7.14MM	0	0	
3180			FLANGE	9001455	16	18	FLG WN 300# RF LTCS A694 F60 WT 7.92MM	0	0	
3190			FLANGE	9001457	20	20	FLG WN 300# RF LTCS A694 F60 WT 8.74MM	0	0	
3200			FLANGE	9001458	24	24	FLG WN 300# RF LTCS A694 F60 WT 9.53MM	0	0	
3210	U04	821	FLANGE	9001454	8	10	FLG WN 300# RF LTCS A694 F60 WT 11.13MM	0	0	
3220	U04	821	FLANGE	9001467	8	16	FLG WN 300# RF LTCS A694 F60 WT 12.7MM	0	0	
3230	U04	821		9006231	16	16	FLG WN 300# RF LTCS A694 F60 WT 15.88MM	0	0	
3240	U04	821	FLANGE	9006230	18	18	FLG WN 300# RF LTCS A694 F60 WT 17.48MM	0	0	
3250	U04	821		9006234	8	8	FLG WN 300# RF LTCS A694 F60 WT 18.26MM	0	0	
3260	U04	821	FLANGE	9006229	20	20	FLG WN 300# RF LTCS A694 F60 WT 19.05MM	0	0	
3270	U04	821	FLANGE	9006229	18	18	FLG WN 300# RF LTCS A694 F60 WT 19.05MM	0	0	
3280	U04	821	FLANGE	9006228	20	24	FLG WN 300# RF LTCS A694 F60 WT 20.62MM	0	0	
3290	U04	821	FLANGE	9006233	24	24	FLG WN 300# RF LTCS A694 F60 WT 20.62MM, TAPERED AT WELDED END TO 9.53 MM	0	0	
3300	U04	821	FLANGE	9003441	10	10	FLG WN 300# RF LTCS A694 F60 WT 21.43MM	0	0	
3310	U04	821	FLANGE	9003442	6	8	FLG WN 300# RF LTCS A694 F60 WT SCH 120	0	0	
3320	U04		BLIND	5615576	0.75	3	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
3330			BLIND	9001450	4	24	FLG BLIND 300# RF LTCS A694 F60	0	0	
3340	U04	FS1		9001518	16	24	FLG BLIND 600# RF LTCS A694 F60	0	0	
3350			FLANGE	9001459	28	28	FLG WN 300# RF LTCS A694 F60 WT 10.31MM SERIES A	0	0	
3360			BLIND	9001451	28	28	FLG BLIND 300# RF LTCS A694 F60 SERIES A	0	0	
3370	U04	FS1	FLANGE	5557036	1	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3380		FS1	FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
3390		FS1	FLANGE	9001471	6	10	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0	
3400		FS1	FLANGE	9003493	12	14	FLG WN 600# RF LTCS A694 F60 WT 7.14 MM	0	0	
3410		FS1	FLANGE	9003494	16	16	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM	0	0	
3420		FS1	FLANGE	9003485	24	24	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM	0	0	
3430	U04	FS1	FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3440	U04	821	FLANGE	9001512	16	16	FLG WN 600# RF LTCS A694 F60 WT 12.70MM	0	0	
3450	U04	821		9005128	16	16	FLG WN 600# RF LTCS A694 F60 WT 15.88 MM	0	0	
3460	U04	FS1		9001517	24	24	FLG WN 600# RF LTCS A694 F60 WT 20.62MM	0	0	
3470	U04	FS1	FLANGE	9003484	20	20	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0	
3480	U04	FS1	FLANGE	5555935	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3490	U04	FS1	FLANGE	9006755	6	10	FLG WN 900# RF LTCS A694 F60 WT 6.35MM	0	0	
3500	U04	FS1,	FLANGE	9006757	12	12	FLG WN 600# RTJ A694 F60 WT 7.14MM	0	0	
		817								
3510	U04	FS1,	FLANGE	9006711	18	18	FLG WN 600# RTJ A694 F60 WT 7.92MM	0	0	
		817								
3520	U04	FS1	FLANGE	5557393	2	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
3530	U04		TEE RED	5599703	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	
3540	U04		FLANGE	9006752	3	3	FLG WN REDUCING 300# RF LTCS A694 F60 BORED TO SUIT WT 6.02MM	4	4	
3550										
3550	U04		ORIFICE FLG	5623944	3	3	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
3560										
3560			FLANGE	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
3570										
3570							***** GASKETS *****			
3580										
3580			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3590			GASKET	5674032	28	28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
3600	U04	FS1	GASKET	5673103	1	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3610	U04	FS1,	GASKET	5673202	12	18	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
		817								
3620	U04	FS1	GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3630	U04	FS1	GASKET	5673117	3	10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3640										
3640		B1					***** BOLTING *****			
3650										
3650	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
3660	U04	821		5675776	0.5	24	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
3670										
3670		SB6	BOLTING	5675675	12	28	JACK SCREW HEX HEAD A320 GR L7M	0	0	
3680										
3680			SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
3690	U04	817	SPEC BLIND	5749587	1	1	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
3700										
3700			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
3710			PADDLE	5750028	28	28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3720			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
3730			PADDLE	5750198	28	28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3740										

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 6 of 6	300K5A